

Prelab 9

April 3, 2026

Prelab 9a: Charge to Mass Ratio

Why is it important to measure charge-to-mass ratio (e/m)?

Measuring the charge-to-mass ratio e/m of the electron is important because the charge e and the mass m are difficult to measure independently. The ratio e/m can be used to determine the motion of charged particles in electric and magnetic fields without needing to know that individual values of e and m .

Prelab 9b: Magnetic Force on a Wire

One consequence of placing a current carrying wire in a magnetic field is that the wire experiences a force $\vec{F} = I \vec{L} \times \vec{B}$ on it, where I is the current, L is the length of the wire within the magnetic field, and $\vec{B}$ is the magnetic field vector.

Note the use of the cross product, which both modifies the magnitude of the force, and defines its direction. If I , L , and B are expressed in units of Amps, meters, Teslas respectively, then the force is expressed in Newtons.

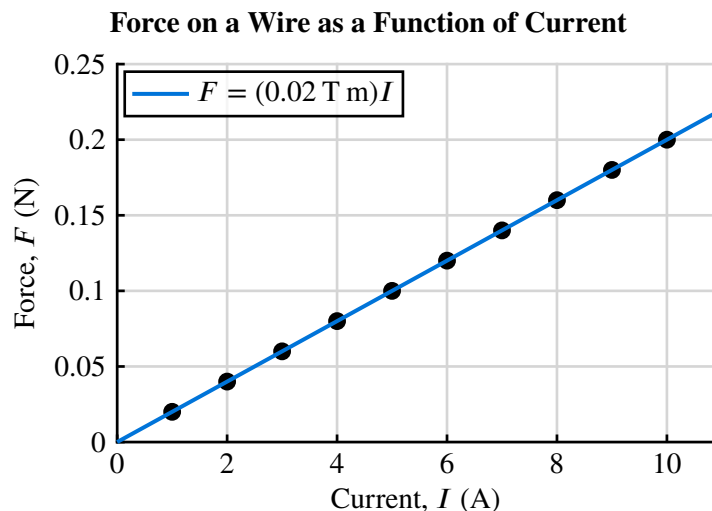
1. Make a properly labeled plot of the Force on a wire of length of wire 1 cm long that is perpendicular to a magnetic field of 2 T, as the current ranges from 1 A to 10 A. What is the slope of this plot? What does that slope represent?

Since the wire is perpendicular to the field, $\sin \theta = 1$, and the force simplifies to:

$$F = ILB$$

With $L = 1 \text{ cm} = 0.01 \text{ m}$ and $B = 2 \text{ T}$:

$$F = I \cdot 0.01 \text{ m} \cdot 2 \text{ T} = (0.02 \text{ T m})I$$



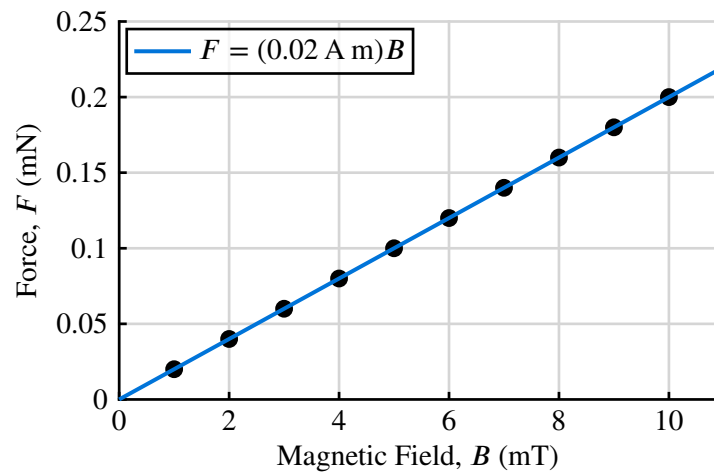
The slope of this plot is 0.02 T m. It represents the product LB , i.e. the length of the wire multiplied by the magnetic field strength.

2. Make a properly labeled plot of the Force on a wire of length of wire 1 cm long with a current of 2 A. The wire is perpendicular to a magnetic field that ranges from 1 mT to 10 mT. What is the slope of this plot? What does that slope represent?

With $L = 0.01$ m and $I = 2$ A:

$$F = ILB = 2 \text{ A} \cdot 0.01 \text{ m} \cdot B = (0.02 \text{ A m})B$$

Force on a Wire as a Function of Magnetic Field Strength



The slope of this plot is 0.02 A m. This slope represents the product IL , i.e. the current through the wire multiplied by the length of the wire.